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COMP3751 VR Coursework Report

PROBLEM 6.1 Static renders and testing evidence:

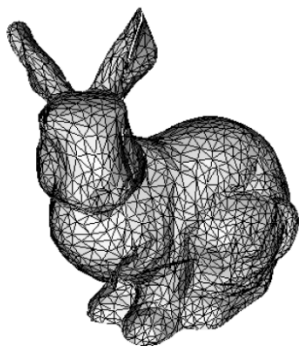


Figure 1: Bunny rendered at its initial orientation, demonstrating the baseline rendering with flat shading and outline edges.

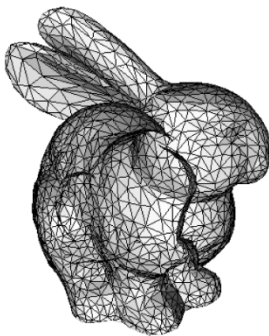


Figure 2: Bunny rotated by IMU gyroscope data. This demonstrates that the rotation transformation matrices are functioning correctly

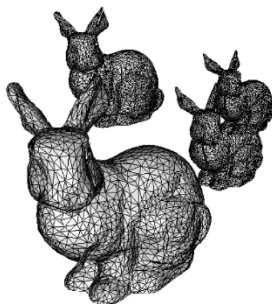


Figure 3: Physics simulation showing the central IMU-driven bunny with multiple falling bunnies. This demonstrates multi-model rendering, translation and scaling transformations.

PROBLEM 3.3 Testing different alpha values:

Setting: samples_per_frame = 100

Frame 10 = IMU [900]

Yaw difference

12.2°(Gyroscope Only) -> 12.1°(alpha =0.01)-> 12.0°(alpha =0.02)

-> 11.9° (alpha =0.05) -> 11.9° (alpha =0.1)

	Gyroscope Only	alpha =0.01	alpha =0.02	alpha =0.05	alpha =0.1
Frame 10	roll=49.5° pitch= -2.0°	roll=46.9° pitch=-2.9°	roll=47.1° pitch=-2.8°	roll=48.1° pitch=-2.4°	roll=49.0° pitch=-2.3°
Frame 20	roll=-91.2° pitch=-2.2°	roll=-94.0° pitch=0.6°	roll=-93.9° pitch=1.0°	roll=-93.9° pitch=1.0°	roll=-93.8° pitch=0.9°
Frame 30	roll=5.2° pitch=68.7°	roll=1.3° pitch=67.9°	roll=2.0° pitch=68.2°	roll=2.6° pitch=68.8°	roll=2.3° pitch=69.0°
Frame 40	roll=-155.6° pitch=-89.1°	roll=177.1 pitch=-88.0°	roll=176.4° pitch=-88.1	roll=175.0° pitch=-88.3°	roll=169.3° pitch=-88.6°
Frame 50	roll=-0.6° pitch=-0.4°	roll=0.5° pitch=2.8°	roll=0.6° pitch=3.0°	roll=0.2° pitch=3.0°	roll=-0.1° pitch=2.8°
Frame 60	roll=-15.2° pitch=5.2°	roll=-17.0° pitch=2.9°	roll=-16.9° pitch=3.4°	roll=-16.5° pitch=4.4°	roll=-16.3° pitch=5.6°
Frame 70	roll=-1.2° pitch=1.3°	roll=-3.3° pitch=1.1	roll=-3.4° pitch=1.1°	roll=-3.7° pitch=1.2°	roll=-4.0° pitch=1.3°

To evaluate the effect of the accelerometer complementary filter coefficient, alpha values of 0.01, 0.02, 0.05, and 0.1 were tested, using gyroscope-only dead reckoning as the baseline.

As expected, the accelerometer correction has minimal impact on yaw, since gravity acts vertically and does not provide heading information. For pitch, a modest improvement is observed across all alpha values, with the error decreasing from 2.0° in the gyroscope-only case to a range between 1.1° and 1.9°.

In contrast, the roll error increases when accelerometer correction is applied, rising from -2.2° to approximately -4.2°. This can be attributed to the fact that, during rapid rotational motion, the accelerometer measures both gravitational acceleration and motion-induced centripetal acceleration. Consequently, the correction may bias the estimated orientation in an incorrect direction. This behaviour is consistent across all tested alpha values, indicating that it is primarily caused by the nature of the motion sequence rather than the choice of filter parameter.

Examining the mid-sequence behaviour, the differences between alpha values become more pronounced during periods of active rotation. At Frame 40, where the pitch is approximately -90°, the corresponding roll values are 177.1°, 176.4°, 175.0°, and 169.3° for alpha values of

0.01, 0.02, 0.05, and 0.1, respectively. Larger alpha values result in greater deviation at this point, suggesting that stronger accelerometer correction introduces increased disturbance when the acceleration measurement is affected by non-gravitational forces.

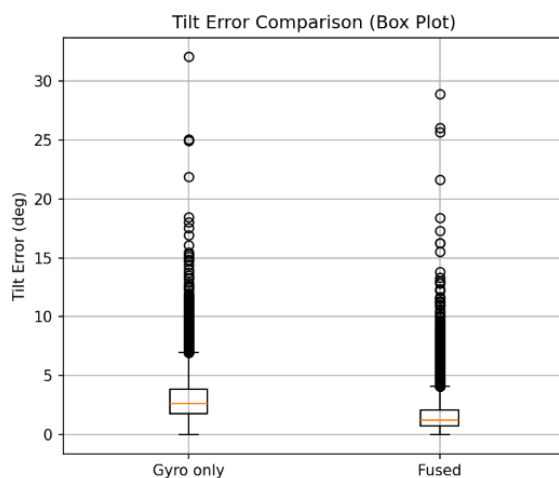
PROBLEM 6.2 Dead reckoning vs accelerometer:

To compare the two methods, tilt error is used as the primary metric rather than examining roll, pitch, and yaw separately. Tilt error is the angle between the estimated gravity direction and the global up vector. This metric captures how well the estimated orientation aligns with gravity, which is particularly relevant for evaluating roll and pitch accuracy.

By reducing each frame's three-dimensional orientation into a single scalar value, tilt error enables straightforward statistical comparison across the entire sequence using metrics such as mean, median, and percentiles. This avoids the complexity of analysing individual Euler angles and provides a more robust and interpretable measure of overall orientation accuracy.

	Gyro only	Gyro +accel ($\alpha=0.02$)	changed
Mean	3.01°	1.73°	42% ↓
Median	2.66°	1.26°	53% ↓
P95	6.24°	4.71°	25% ↓
Max	32.04°	28.87°	10% ↓

Table: Tilt Error Statistics for Gyroscope-Only vs Accelerometer-Fused Tracking



The box plot shows that the fused method has a lower median and a narrower interquartile range, indicating that both the typical error and the variability of the error are reduced.

Simple dead reckoning has the advantage of simplicity and low computational cost. Additionally, since it does not rely on acceleration measurements, it avoids introducing noise during rapid motion. However, its main disadvantage is the continuous accumulation of drift, as there is no absolute reference. This results in increasing error over time, with a mean tilt error of 3.01 degrees over the 27-second sequence.

Including the accelerometer significantly improves accuracy by introducing a gravity-based reference. The fused method reduces the mean tilt error by 42% and the median by 53%, demonstrating improved overall performance and stability. The gravity reference constrains roll and pitch, preventing unbounded drift. However, this comes at the cost of increased

computational complexity and sensitivity to noise. During rapid motion, acceleration measurements deviate from pure gravity, which reduces the effectiveness of the correction. This is reflected in the maximum tilt error, which only decreases by 10%, indicating limited improvement in highly dynamic segments. Although some individual frames show increased roll deviation under accelerometer correction, the overall tilt-error statistics across the full sequence indicate that accelerometer fusion improves roll/pitch stability on average

Overall, dead reckoning is simple and efficient but suffers from drift, while incorporating accelerometer data improves accuracy and stability but introduces noise sensitivity and higher computational cost.

PROBLEM 6.3 Performance comparison:

Dead reckoning has low computational cost, as it only requires quaternion integration based on angular velocity measurements. Each update involves computing the rotation axis and angle, followed by a single quaternion multiplication. These operations are relatively simple and efficient, making the method well-suited for real-time applications. The dead reckoning method runs faster than the fused method in practice, due to the lower computational cost per update.

In contrast, the fused method (gyro + accelerometer) introduces several additional operations. These include transforming the acceleration vector into the global frame using quaternion rotation, normalising the acceleration vector, computing cross and dot products to determine the tilt axis and angle, evaluating trigonometric functions, and applying an additional quaternion correction step. These extra steps significantly increase the number of computations required per update.

As a result, the fused method requires significantly more computations per update compared to dead reckoning, leading to higher computational cost and reduced runtime performance.

PROBLEM 4.2 Testing different alpha_mag values:

Setting: samples_per_frame = 100 alpha=0.02

Frame 10 = IMU [900]

	alpha_mag=0	alpha_mag=0.01	alpha_mag=0.02	alpha_mag=0.05	alpha_mag=0.1
Frame 10	yaw=-3.0°	yaw=-3.3°	yaw=-2.7	yaw=-1.0°	yaw=-0.1
Frame 20	yaw=-8.6°	yaw=-2.1°	yaw=0.1°	yaw=0.3°	yaw=0.2
Frame 30	yaw=0.7°	yaw=-0.9°	yaw=-0.7	yaw=-0.6°	yaw=-0.5
Frame 40	yaw=-166.0°	yaw=15.5°	yaw=4.0°	yaw=1.1	yaw=0.9
Frame 50	yaw=112.1°	yaw=25.3°	yaw=7.0°	yaw=0.2°	yaw=-0.2

Frame 60	yaw=-19.2°	yaw=39.5°	yaw=27.2°	yaw=11.9°	yaw=5.4
Frame 70	yaw=11.9°	yaw=-0.1	yaw=-0.0°	yaw=0.1°	yaw=0.1

In the full fusion experiment, I fixed the tilt-correction parameter at $\alpha = 0.02$ and tested different magnetometer complementary-filter coefficients, α_{mag} in $\{0.01, 0.02, 0.05, 0.1\}$. It should be noted that the yaw at intermediate time steps is not necessarily expected to be 0, since the object may undergo genuine heading changes during the motion sequence.

Nevertheless, the overall trend in the results shows that the choice of α_{mag} has a clear effect on yaw drift compensation. As α_{mag} increases, the magnetometer contributes more strongly to yaw correction, and is therefore able to suppress heading drift caused by gyroscope integration more effectively. This trend is particularly evident in the later part of the sequence. For example, at Frame 60, the yaw values were 39.5, 27.2, 11.9, and 5.4 degrees respectively; at Frame 50, they were 25.3, 7.0, 0.2, and -0.2 degrees. This indicates that smaller values of α_{mag} provide insufficient correction, whereas larger values significantly reduce the residual yaw deviation in the later stages of the sequence. This suggests that part of the observed yaw deviation is caused by gyroscope drift, rather than being entirely due to normal pose changes.

However, larger α_{mag} values make the system more susceptible to magnetic interference from nearby metal objects or electronic devices, while smaller values offer better noise resistance at the cost of slower drift correction.

In addition to the standard tilt-compensated heading computation, a simple magnetometer-magnitude check was added as a robustness measure. Yaw correction was only applied when the measured magnetic-field magnitude remained within 30% of a reference value, so that potentially disturbed magnetometer readings would not introduce unstable heading updates.

PROBLEM 6.4 Positional tracking from IMU:

We could use IMU data to track position by integrating acceleration twice. The accelerometer measures linear acceleration, so integrating it once gives velocity, and integrating it again gives position.

However, the main problem is that even very small errors in the acceleration readings build up over time. Because position comes from integrating twice, these errors grow really quickly. On top of that, if the orientation isn't estimated accurately, the system can't properly remove the effect of gravity, which introduces even more error.

PROBLEM 6.5 Sphere-based collision detection:

Our implementation uses sphere-based collision detection, where each bunny is enclosed by a bounding sphere. A collision is detected when the distance between two centres is less than the sum of their radius. This is efficient since it only requires one distance calculation per object pair.

The main limitation is that the bunny model is not spherical. The bounding sphere includes significant empty space around the ears and legs, causing collisions to be detected when objects are still visually separated. For the provided bunny model with 2503 vertices and 4968 triangles, this inaccuracy is particularly noticeable due to the elongated shape of the model.

First, axis-aligned bounding boxes or oriented bounding boxes can be used instead of a single bounding sphere. These provide a tighter fit around the model, reducing false positive collisions, although they introduce slightly higher computational cost compared to simple sphere-based distance checks.

Second, multiple spheres can be used to approximate the bunny model. For example, separate spheres can be assigned to the body, ear and head. This approach remains significantly faster than triangle-level collision detection while providing much higher accuracy than a single sphere